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LINEAR ALGEBRA • UNIT 8.01

Orthogonality, Gram-Schmidt, QR & Least Squares

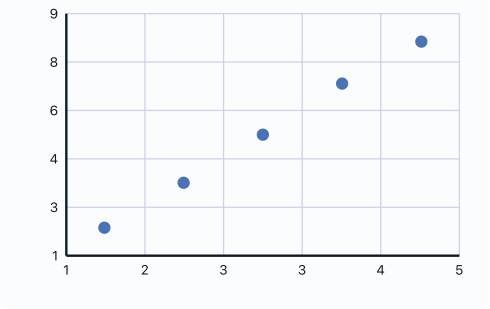
PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Orthogonality, Gram-Schmidt, QR & Least Squares Practice Exercises

1 MCQ

For which k are $\langle 0, -2, -2\rangle$ and $\langle -1, 2, k\rangle$ orthogonal?



A 2

B -2

C -1

D 3

2 MCQ

Which set is NOT orthonormal?

A $\{\langle \frac{5}{13}, \frac{12}{13}\rangle, \langle \frac{12}{13}, -\frac{5}{13}\rangle\}$

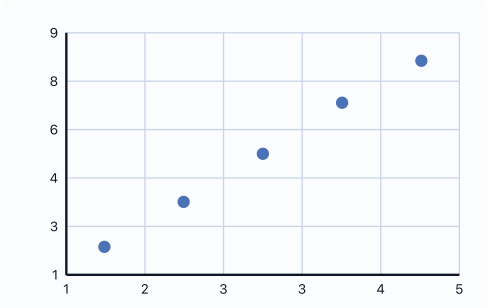
B $\{\langle \frac{1}{2}, \frac{1}{2}\rangle, \langle \frac{1}{2}, -\frac{1}{2}\rangle\}$

C $\{\langle \frac{1}{3}, \frac{2}{3}, \frac{2}{3}\rangle, \langle \frac{2}{3}, \frac{1}{3}, -\frac{2}{3}\rangle\}$

D $\{\langle \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\rangle, \langle \frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\rangle\}$

3 MCQ

Which matrix is orthogonal (satisfies $Q^T Q = I$)?



A $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

B $\begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix}$

C $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

D $\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$

4 MCQ

Normalize $\mathbf{v} = \langle 6, 8 \rangle$.

A $\langle \frac{3}{5}, \frac{4}{5} \rangle$

B $\langle \frac{3}{50}, \frac{2}{25} \rangle$

C $\langle \frac{3}{7}, \frac{4}{7} \rangle$

D $\langle -\frac{3}{5}, -\frac{4}{5} \rangle$

5 MCQ

Which set is orthonormal?

A $\{\langle 1, 2, 2 \rangle, \langle 2, 1, -2 \rangle\}$

B $\{\langle \frac{1}{3}, \frac{2}{3}, \frac{2}{3} \rangle, \langle \frac{2}{3}, \frac{1}{3}, -\frac{2}{3} \rangle\}$

C $\{\langle \frac{1}{2}, \frac{1}{2} \rangle, \langle \frac{1}{2}, -\frac{1}{2} \rangle\}$

D $\{\langle \frac{3}{5}, \frac{4}{5} \rangle, \langle \frac{4}{5}, \frac{3}{5} \rangle\}$

6 MCQ

For which k are $\langle 3, 3, 2 \rangle$ and $\langle 2, k, 1 \rangle$ orthogonal?

A $\frac{8}{3}$

B $-\frac{5}{3}$

C $\frac{11}{3}$

D $-\frac{8}{3}$

7 MCQ

Normalize $\mathbf{v} = \langle -2, 1, -2 \rangle$.

A $\langle -\frac{2}{9}, \frac{1}{9}, -\frac{2}{9} \rangle$

B $\langle -\frac{2}{5}, \frac{1}{5}, -\frac{2}{5} \rangle$

C $\langle \frac{2}{3}, -\frac{1}{3}, \frac{2}{3} \rangle$

D $\langle -\frac{2}{3}, \frac{1}{3}, -\frac{2}{3} \rangle$

8 MCQ

For which k are $\langle 0, -3, 1 \rangle$ and $\langle 1, k, -1 \rangle$ orthogonal?

A $\frac{1}{3}$

B $\frac{2}{3}$

C $-\frac{1}{3}$

D $\frac{4}{3}$

9 MCQ

Consider two vectors $\mathbf{v}_1 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} -4 \\ 3 \end{pmatrix}$. What is the dot product $\mathbf{v}_1 \cdot \mathbf{v}_2$?

A 1

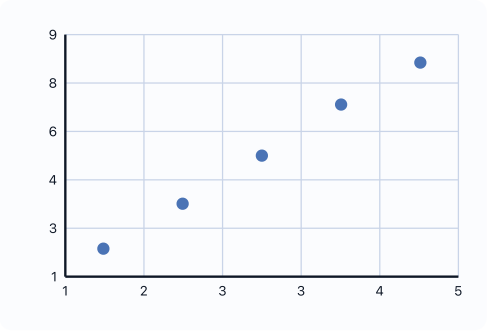
B -7

C 0

D 25

10 MCQ

Consider the matrix $A = \begin{pmatrix} 1 & 2 \\ -2 & k \end{pmatrix}$. If the column vectors of A are orthogonal, find the value of k.



A 4

B -2

C -1

D 2

11 MCQ

Consider the matrix $A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$. Let $\mathbf{b} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$. We want to find the vector $\hat{\mathbf{x}}$ that minimizes $\|A\mathbf{x} - \mathbf{b}\|$. This is equivalent to finding the projection of $\mathbf{b}$ onto the column space of A, denoted by $\text{Col}(A)$. The normal equations for this least squares problem are $A^T A\mathbf{x} = A^T\mathbf{b}$. The unique solution $\hat{\mathbf{x}}$ to the normal equations yields the best approximate solution. Calculate the value of $\hat{\mathbf{x}}_1$, the first component of $\hat{\mathbf{x}}$.

A 1/3

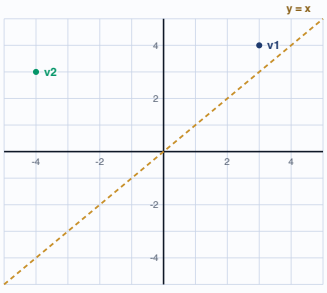
B 2/3

C 1/2

D 3/4

12 MCQ

Consider the vectors $\mathbf{v}_1 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} -4 \\ 3 \end{pmatrix}$. What is the dot product $\mathbf{v}_1 \cdot \mathbf{v}_2$?



A 0

B 1

C 7

D -7

13 MCQ

A 3×3 symmetric matrix T_a has eigenvalues $\lambda_1 = 9$, $\lambda_2 = 24$, and $\lambda_3 = 20$. Determine the trace $\text{tr}(T_a)$.

A 29

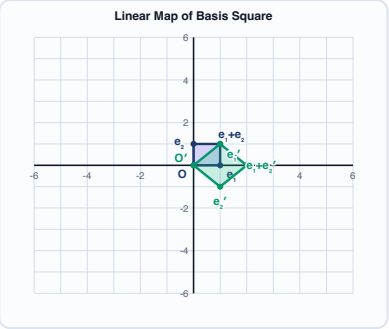
B 53

C 56

D 62

14 MCQ

Consider the matrix $A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$. Let $\mathbf{v} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$. Suppose we want to find the vector $\mathbf{x}$ of the form $c_1\mathbf{a}_1 + c_2\mathbf{a}_2$ where $\{\mathbf{a}_1, \mathbf{a}_2\}$ is an orthogonal basis derived from the columns of A, such that $\|\mathbf{v} - \mathbf{x}\|$ is minimized. Find the value of $\|\mathbf{x}\|$.



A $2\sqrt{10}$

B $\sqrt{20}$

C $\sqrt{5}$

D $\sqrt{10}$

15 MCQ

Consider the vector space R^2 with the standard dot product. Let $\mathbf{u} = (2, 3)$ and $\mathbf{v} = (-3, 2)$. Which of the following statements about $\mathbf{u}$ and $\mathbf{v}$ is true?

A $\mathbf{u}$ and $\mathbf{v}$ are linearly dependent.

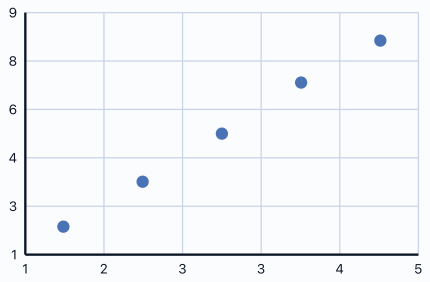
B $\mathbf{u}$ and $\mathbf{v}$ are orthogonal.

C The angle between $\mathbf{u}$ and $\mathbf{v}$ is acute.

D The angle between $\mathbf{u}$ and $\mathbf{v}$ is obtuse.

16 MCQ

Apply Gram–Schmidt to $\mathbf{x}_1 = \langle 1, -1 \rangle$, $\mathbf{x}_2 = \langle 2, 3 \rangle$. Which vector is parallel to the resulting $\mathbf{v}_2$?



A $\langle 2, 3 \rangle$

B $\langle \frac{3}{2}, \frac{7}{2} \rangle$

C $\langle 1, -1 \rangle$

D $\langle 1, 1 \rangle$

17 MCQ

Apply Gram–Schmidt to $\mathbf{x}_1 = \langle 3, 2 \rangle$, $\mathbf{x}_2 = \langle 0, 2 \rangle$. Which vector is parallel to the resulting $\mathbf{v}_2$?

A $\langle -2, 3 \rangle$

B $\langle 0, 2 \rangle$

C $\langle \frac{12}{13}, \frac{34}{13} \rangle$

D $\langle 3, 2 \rangle$

18 MCQ

Apply Gram–Schmidt to $\mathbf{x}_1 = \langle 1, 1 \rangle$, $\mathbf{x}_2 = \langle 2, -2 \rangle$. Which vector is parallel to the resulting $\mathbf{v}_2$?

A $\langle -1, 1 \rangle$

B $\langle 1, -1 \rangle$

C $\langle 2, -2 \rangle$

D $\langle 1, 1 \rangle$

19 MCQ

Apply Gram–Schmidt to $\mathbf{x}_1 = \langle -1, 3, 1 \rangle$, $\mathbf{x}_2 = \langle 3, -3, -2 \rangle$. Which vector is parallel to the resulting $\mathbf{v}_2$?

A $\langle -1, 3, 1 \rangle$

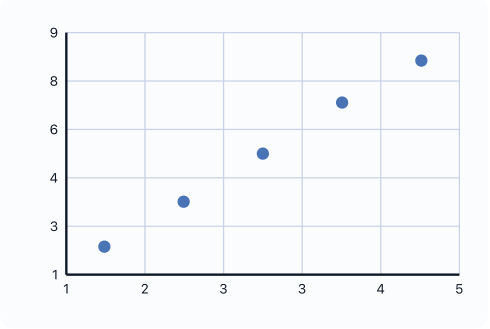
B $\langle 3, -3, -2 \rangle$

C $\langle 19, 9, -8 \rangle$

D $\langle \frac{47}{11}, -\frac{75}{11}, -\frac{36}{11} \rangle$

20 MCQ

Apply Gram–Schmidt to $\mathbf{x}_1 = \langle -1, 3\rangle$, $\mathbf{x}_2 = \langle -3, 2\rangle$. Which vector is parallel to the resulting $\mathbf{v}_2$?



A $\langle -3, -1\rangle$

B $\langle -3, 2\rangle$

C $\langle -\frac{39}{10}, \frac{47}{10}\rangle$

D $\langle -1, 3\rangle$

21 MCQ

Apply Gram–Schmidt to $\mathbf{x}_1 = \langle 2, -1, 0\rangle$, $\mathbf{x}_2 = \langle 0, 3, -1\rangle$. Which vector is parallel to the resulting $\mathbf{v}_2$?

A $\langle 0, 3, -1\rangle$

B $\langle -\frac{6}{5}, \frac{18}{5}, -1\rangle$

C $\langle 2, -1, 0\rangle$

D $\langle 6, 12, -5\rangle$

22 MCQ

Apply Gram–Schmidt to $\mathbf{x}_1 = \langle -2, 3, 0\rangle$, $\mathbf{x}_2 = \langle 1, 3, 2\rangle$. Which vector is parallel to the resulting $\mathbf{v}_2$?

A $\langle 1, 3, 2\rangle$

B $\langle -\frac{1}{13}, \frac{60}{13}, 2\rangle$

C $\langle 27, 18, 26\rangle$

D $\langle -2, 3, 0\rangle$

23 MCQ

Apply Gram–Schmidt to $\mathbf{x}_1 = \langle -2, 2\rangle$, $\mathbf{x}_2 = \langle -3, 2\rangle$. Which vector is parallel to the resulting $\mathbf{v}_2$?

A $\langle -3, 2\rangle$

B $\langle -\frac{11}{2}, \frac{9}{2}\rangle$

C $\langle -1, -1\rangle$

D $\langle -2, 2\rangle$

24 MCQ

Apply Gram–Schmidt to $\mathbf{x}_1 = \langle 3, 3\rangle$, $\mathbf{x}_2 = \langle -2, -1\rangle$. Which vector is parallel to the resulting $\mathbf{v}_2$?

A $\langle -1, 1\rangle$

B $\langle 3, 3\rangle$

C $\langle -2, -1\rangle$

D $\langle -\frac{7}{2}, -\frac{5}{2}\rangle$

25 MCQ

Apply Gram–Schmidt to $\mathbf{x}_1 = \langle 3, 3\rangle$, $\mathbf{x}_2 = \langle -3, 1\rangle$. Which vector is parallel to the resulting $\mathbf{v}_2$?

A $\langle 3, 3\rangle$

B $\langle -3, 1\rangle$

C $\langle -1, 1\rangle$

D $\langle -4, 0\rangle$